higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

# MARKING GUIDELINE

## NATIONAL CERTIFICATE (VOCATIONAL)

## MATHEMATICAL LITERACY
## (Second Paper)
## NQF LEVEL 3

## 6 November 2019

| SYMBOL | EXPLANATION |
|---|---|
| M | Method |
| MA | Method with accuracy |
| CA | Consistent accuracy |
| A | Accuracy |
| C | Conversion |
| S | Simplification |
| RT/RG/RD/RM | Reading from a table/graph/drawing/document/map |
| F | Choosing correct formula |
| SF | Substitute into formula |
| R/J | Reasoning/Justification |
| P | Penalty – no units, incorrect rounding off |
| R | Rounding off |
| E | Explanation |
| U | Unit |
| MF | Manipulation of Formula |

**This marking guideline consists of 9 pages.**

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| **QUESTION 1 [40] *Do not penalise if units are omitted unless marks are allocated.** | | |
|---|---|---|
| **Question** | **Solution** | **Explanation** |
| 1.1<br>1.1.1 | Radius = 9✓ cm<br>$\text{Volume} = \frac{1}{3}\pi r^2 h$<br>$= \frac{1}{3} \times 3{,}14\checkmark \times 9^2\checkmark \times 24\checkmark$<br>$= 2\,034\checkmark \text{ cm}^3\checkmark$ | 1 A<br><br><br>3 SF (correct formula)<br><br>1 A 1 U<br>(6) |
| 1.1.2 | Surface area = $\pi$ r $\sqrt{r^2+h^2}$<br>$= 3{,}14 \times 9\checkmark \times \sqrt{9^2\checkmark + 24^2\checkmark}$<br>$= 724{,}36\checkmark \text{ cm}^2\checkmark$ | <br><br>3 SF<br>1 A<br>(5) |
| 1.1.3 | $37\,500 = \frac{1}{2} \times \text{ BC} \times 250\checkmark$<br>$37\,500 = 125 \times \text{ BC}\checkmark$<br>$\therefore \text{BC} = \frac{37\,500\checkmark}{125\checkmark}$<br>$= 300\text{mm}$<br>Or<br>BC ✓$= \frac{2\checkmark \times 37\,500\checkmark}{250\checkmark}$<br>$= 300$ mm | 1 SF<br><br>1M<br><br>2 MF<br><br><br>2 MF<br>1 M<br>1 SF<br>(4) |
| 1.1.4 | DC = 300 ÷ 2 = 150 mm✓ | 1A<br>(1) |
| 1.1.5 | $\text{AC}^2 = \text{AD}^2 + \text{DC}^2$<br>$= 250^2 + 150^2\checkmark$<br>$= 85\,000\checkmark$<br>$\therefore \text{AD} = \sqrt{85\,000}\ \checkmark$<br>$= 291{,}547\checkmark$ mm<br>$= 292\ \checkmark$mm | <br><br><br>1 SF<br>1 CA<br><br>1 M<br>1 CA<br>1 R<br>(5) |
| 1.1.6 | Edging = 300✓ + 292✓ + 292✓<br>= 884✓ mm<br>= 0,884✓ m | (CA Q1.1.5)<br>3 M<br>1 CA<br>1 C<br>(5) |

| *Do not penalise if units are omitted unless marks are allocated. | | |
|---|---|---|
| **Question** | **Solution** | **Explanation** |
| 1.2<br>1.2.1 | A5 ✓and E1 ✓ | 2 RG<br>(2) |
| 1.2.2 | Boundary Street ✓and Hospital Street ✓ | 2 RG<br>(2) |
| 1.2.3 | 145 mm✓✓ | 2 A (Please note that the answer might differ based on final print of the Question Paper)<br>(2) |
| 1.2.4 | Actual Distance in mm = 3,5 × 1 000 000✓<br>= 3 500 000✓<br>Map distance = 3 500 000 ÷ 20 000✓<br>= 175✓mm | 1 M × 1 000 000<br>1 C<br>1 M ÷ 20 000<br>1 CA<br>(4) |
| 1.2.5 | Time = $\frac{8}{60}$ = 0.1333 hours✓<br>Speed = $\frac{3,9✓}{0,1333✓}$<br>= 29,25✓km/h | 1 C 0,1333 hours<br>2 SF<br>1 CA<br>(4) |

| QUESTION 2 [40] *Do not penalise if 'R' is omitted. | | |
|---|---|---|
| **Question** | **Solution** | **Explanation** |
| 2.1<br>2.1.1 | The total amount an employee will be paid before tax✓ and deductions. ✓<br><br>Or<br><br>The total salary package of an employee. ✓ It indicates the total amount of expenses an employer spends on an employee✓ per month or per year. | 2E<br>(2) |
| 2.1.2 | R29 000 ✓ × (19 ÷ 100) ✓<br>= R5 510✓ | 2 M<br>1 A<br>(3) |
| 2.1.3 | Retirement fund (ABC Electronics) = R20 000 ✓ x 5%✓<br>= R1 000✓ | 1 RT 1 M<br>1 A<br>(3) |

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|---|---|---|
| 2.1.4 | Medical aid cost = R2 130✓ + R 1 370✓<br>= R3 500✓ | 1 RT 1 M<br>1 A<br>(3) |
| 2.1.5 | Retirement fund: R1 000 + R1 250✓<br>= R2 250✓ | (CA 2.1.3)<br>1 M<br>1 CA<br>(2) |
| 2.1.6 | Magic Electronics offer a Total package R29 000. This amount includes benefits, overtime allowances. ✓ Anna would, therefore have to pay all deductions from this amount and she would lose all her benefits. ✓<br><br>(Any other relevant explanation) | 2 E<br>(2) |
| 2.2<br>2.2.1 | • Electricity✓<br>• Water ✓<br>• Transport<br><br>(Any 2 relevant expenses) | 2A<br>(2) |
| 2.2.2 | Total amount for 12 muffins:<br><br>Muffin mix = 12,00 ÷ 1 000 × 500 ✓<br>= R6,00✓<br><br>Eggs = 10,50 ÷ 6 ✓<br>= R1,75✓<br><br>Oil = 12,50 ÷ 750 × 125 ✓<br>= R2,08 ✓<br><br>Milk = 11,50 ÷ 1 000 × 375 ✓<br>= R4,31✓<br><br>Total for 12 muffins = 6 + 1,75 + 2,08 + 4,31<br>= 14,14 ✓<br><br>Total for 24 muffins = 14,14 × 2<br>= R28,28 ✓<br><br>Or<br><br>Total amount for 24 muffins:<br><br>Muffin mix = 12,00 ÷ 1 000 × 500✓ × 2✓<br>= R12,00✓<br><br>Eggs = 10,50 ÷ 6 ✓× 2<br>= R3,50✓ | <br><br>1 M<br>1 A<br><br>1 M<br>1 A<br><br>1 M<br>1 A<br><br>1 M<br>1 A<br><br><br>1 CA<br><br><br>1 CA<br><br><br><br>2 M<br>1 A<br><br>1 M<br>1 A<br><br>1 M |

| | | |
|---|---|---|
| | Oil = 12,50 ÷ 750 × 125 ✓× 2<br>= R4,16 ✓<br><br>Milk = 11,50 ÷ 1 000 × 375✓ × 2<br>= R8,62✓<br><br>Total for 24 muffins = 12 + 3,50 + 4,16 + 8,62<br>= R28,28 ✓ | 1 A<br><br>1 M<br>1 A<br><br><br>1 CA<br>(10) |
| 2.2.3 | Number of dozens of muffins = 1 200 ÷ 12<br>= 100✓<br>Total cost of ingredients= 14,14 × 100 = R1 414✓ (expenses)<br><br>Total sales = 100 × R30✓ (Income)<br>= R3 000✓<br><br>Profit = R3 000 – R1 414<br>= R1 586✓<br><br>Yes✓ Anna will make a profit of more than R1 000 | <br><br>1 A (100 dozens)<br>1CA (expenses)<br><br>1 M<br>1 A (income)<br><br><br>1 CA (Profit)<br><br>1 A (Yes)<br>(6) |
| 2.2.4 | In 25 minutes she makes 24 muffins:<br>∴ Time taken to bake 240 muffins = 10 dozens × 25 minutes<br>= 250 minutes✓<br><br>She will bake 10 times.<br>Time spent on cleaning pans:<br><br>10 × 10 minutes = 100 minutes✓<br><br>Total time = 250 + 100 = 350 minutes ✓<br><br>Time in hours = 350 ÷ 60<br>= 5,833333333 hours ✓<br><br>= 5 hours 50 minutes✓<br><br>Time muffins will be ready = 09:00 + 6 hours 20 min<br>= 15:20✓<br><br>The muffins will be ready on time✓ | <br><br>1 MA<br><br><br><br>1 A<br><br>1 MA<br><br><br>1 MA<br><br>1 C (hours and minutes)<br><br><br>1 CA<br><br>1A<br>(7) |

| **QUESTION 3 [35] *Do not penalise if 'R' is omitted.** | | |
|---|---|---|
| **Question** | **Solution** | **Explanation** |
| 3.1<br>3.1.1 | 40 000✓✓ | 2 MA<br>(2) |
| 3.1.2 | The 'number of workers' represent natural/whole numbers. Therefore, cannot be a fraction, hence dotted line.✓✓<br><br>Or<br><br>People cannot be represented a fractions. ✓✓ | 2 R<br><br>(2) |
| 3.1.3 | Dependent variable = time taken in days✓<br>Independent variable = number of workers✓ | 1 RG<br>1 RG<br>(2) |
| 3.1.4 | 400 ✓✓workers | 2 RG<br>(2) |
| 3.1.5 | 320✓✓✓days<br><br>Or<br><br>40 000✓ ÷ 125✓<br>= 320 ✓days | 2 RG 1 M (determining correct intervals)<br><br><br>1 M 1 RG<br>1 A<br>(3) |
| 3.2<br>3.2.1 | Taxi fare = R25✓ + 7✓ × number of km ✓<br>(accept other alternatives) | 3 A<br>(3) |
| 3.2.2 | Taxi fare per passenger = R25 + 7 × 12✓<br>= R109✓<br>∴Taxi fare for 3 passengers = R109 × 3✓<br>= R327✓ | (CA 3.2.1)<br>1 SF<br>1 CA<br>1 M<br>1 CA<br>(4) |
| 3.3<br>3.3.1 | R720✓ | 1 RG<br>(1) |
| 3.3.2 | $\text{Percentage increase} = \frac{\text{R1 240 - R720}✓}{\text{R720}✓} \times 100✓$<br>$= 72,22✓\%$ | 3 MA<br><br>1 A<br>(4) |

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| 3.4<br>3.4.1 | Contract 1✓<br>It starts at 0✓<br><br>Or<br><br>As the talk time increase the monthly cost increases in the same proportion | 1 A<br><br>1R<br><br><br><br>(2) |
| 3.4.2 | 100✓ | 1 RG<br>(1) |
| 3.4.3 | 60 minutes✓ and 240 minutes✓ | 2 RG<br>(2) |
| 3.4.4 | R100✓<br>R100✓<br>2,25✓ | 1 RG<br>1 RG<br>1 A<br>(3) |
| 3.4.5 | Contract 2✓<br><br>Difference in monthly cost = R775✓ – 700✓<br>= R75✓ | 1 A<br><br>2 RG and accuracy<br>1 A<br>(4) |

| **QUESTION 4 [35]** | | |
|---|---|---|
| **Question** | **Solution** | **Explanation** |
| 4.1<br>4.1.1 | Mode = 21✓ | 1 RT<br>(1) |
| 4.1.2 | **Vegetarian menu: Median**<br><br>**2 4 5 6 6 9 12 12** ✓<br><br>Median = (6 + 6) ÷ 2<br>= 6 ✓<br><br>**Meat menu: Median**<br><br>**10 11 13 19 20 21 21 21**✓<br><br>Median = (19 + 20) ÷ 2✓<br>= 19,5<br>= 20✓<br><br>Difference = 20 – 6 = 14✓ (Also accept 19,5 – 6 = 13,5) | 1A ascending order<br><br>1A<br><br>1A ascending order<br><br>1M<br><br>1A<br><br>1CA<br>(6) |

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|---|---|---|
| 4.2<br>4.2.1 | Number of females enrolled = 280 + 240 ✓+ 220 + 80✓<br>= 820✓ | 2 RG (1 for every 2 correct)<br>1 CA<br>(3) |
| 4.2.2 | Total students enrolled = 500 + 500 + 420 + 320✓<br>= 1 740✓ | 1 RG (all correct)<br>1 CA<br>(2) |
| 4.2.3 | The Science Faculty | 1 RG<br>(1) |
| 4.2.4 | Males commerce faculty = 420✓ – 220✓<br>= 200✓ | 1 RG 1 M<br>1 A<br>(3) |
| 4.2.5 | Mean Males = 920✓÷ 4✓<br>= 230✓<br><br>Mean Females = 820✓÷ 4<br>= 205✓<br><br>Difference = 230 – 205<br>= 25✓ | 1A sum of 920<br>1M ÷ 4<br>1A<br><br>1CA sum of 820 (Q4.2.1)<br><br>1CA<br><br>1CA<br>(6) |
| 4.2.6 | Enrol more females in the Engineering Faculty✓✓<br><br>Or<br><br>Enrol more males in the Science Faculty/humanities Faculty.✓✓ | 2E<br>(2) |
| 4.3<br>4.3.1 | A = 30✓<br><br>B = 70✓ | 1 A<br><br>1 A<br>(2) |
| 4.3.2 | Probability (still water) $= \frac{40✓}{110✓} = 36,36\%✓$ | 1A numerator<br>1A denominator<br>1CA percentage<br>(3) |

| | | |
|---|---|---|
| 4.3.3 | Expectation $= \frac{40}{110} \times 1200$✓<br>$= 436{,}36$✓<br>$= 436$ ✓customers<br><br>Expectation $= 1200 \times 36.36\%$✓<br>$= 436{,}36$✓<br>$= 436$ ✓customers | 1 M<br>1CA<br>1 R (rounding down)<br><br>1 M<br>1 CA<br>1 R (rounding down)<br>(3) |
| 4.3.4 | Probability (under 20 and sparkling) $= \frac{60✓}{110✓} = \frac{6}{11}$✓ | 1A numerator<br>1A denominator<br>1CA simplified fraction<br>(3) |
| | **TOTAL:** | **150** |

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